CBSE Class 10 Mathematics — Quadratic Equations

Time Allowed: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains $_{questions. All questions are compulsory.}$

1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. Which of the following is a quadratic equation?

(A) (A) $x(x+1)+8=(x+2)(x-2)$
(B) (B) $(x+2)^3=x^3-4$
(C) (C) $(x+2)^2=2(x+3)$
(D) (D) $x^2+\frac{1}{x^2}=5$

2. The discriminant of the quadratic equation $2x^2-4x+3=0$ is:

(A) (A) 8
(B) (B) -8
(C) (C) 4
(D) (D) -4

3. If 1/2 is a root of the equation $x^2-kx-5/4=0$, then the value of k is:

(A) (A) 2
(B) (B) -1
(C) (C) 1
(D) (D) -2

4. The quadratic equation $x^2+kx+1=0$ has equal roots if:

(A) (A) $k=\pm 2$
(B) (B) $k=\pm 1$
(C) (C) $k=0$
(D) (D) $k=\pm 4$

5. The roots of the quadratic equation $x^2-9=0$ are:

(A) (A) 9, -9
(B) (B) 3, 3
(C) (C) 3, -3
(D) (D) 0, 9

6. Assertion (A): The quadratic equation $x^2 + 5x + 6 = 0$ has two distinct real roots. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has distinct real roots if its discriminant $D = b^2 - 4ac > 0$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

7. Assertion (A): If $x = 2$ is a root of $x^2 - kx + 4 = 0$, then $k = 4$. Reason (R): A value $x = \alpha$ is a root of the equation $f(x) = 0$ if $f(\alpha) = 0$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

8. Assertion (A): The equation $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation. Reason (R): Any equation of the form $ax^2 + bx + c = 0$ where $a \neq 0$ is a quadratic equation.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

9. Assertion (A): The quadratic equation $x^2 + 2x + 5 = 0$ has no real roots. Reason (R): For $x^2 + 2x + 5 = 0$, the discriminant is -16.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
(C) (C) Assertion (A) is true but Reason (R) is false
(D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The roots of $x^2 - 9 = 0$ are 3 and -3. Reason (R): $x^2 - a^2 = (x-a)(x+a)$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. Find the roots of the quadratic equation $x^2 - 3x - 10 = 0$ by the method of factorization.

2. Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

3. Check whether $x(x+3) + 1 = (x-2)(x+2)$ is a quadratic equation.

4. Find the roots of the equation $2x^2 - x + \frac{1}{8} = 0$ by using the quadratic formula.

5. The sum of two numbers is 27 and their product is 182. Form the quadratic equation representing this scenario.

Section D: Long Answer (4-5 marks each)

1. Find the roots of the quadratic equation $2x^2 - x + \frac{1}{8} = 0$ by the method of factorization.

2. Find two consecutive positive integers, sum of whose squares is 365.

3. Find the value of k for which the quadratic equation $2x^2 + kx + 3 = 0$ has two equal real roots.

4. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

5. Solve the quadratic equation $x^2 - 5x + 6 = 0$ using the quadratic formula.

6. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

7. The sum of the areas of two squares is 468 m^2. If the difference of their perimeters is 24 m, find the sides of the two squares.

8. A peacock is sitting on the top of a pillar, which is 9 m high. From a point 27 m away from the bottom of the pillar, a snake is coming to its hole at the base of the pillar. Seeing the snake, the peacock pounces on it. If their speeds are equal, at what distance from the hole is the snake caught?

9. A shopkeeper buys a number of books for 80 rupees. If he had bought 4 more books for the same amount, each book would have cost 1 rupee less. How many books did he buy?